

Week 2 Worksheet: Superposition and Quantum Gates - ANSWER KEY

QCA Fundamentals

Probability Amplitudes and Normalization

1. A qubit is in the state $|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle$.

- (a) Probability of $|0\rangle = |\frac{3}{5}|^2 = \frac{9}{25} = 0.36 = 36\%$
- (b) Probability of $|1\rangle = |\frac{4}{5}|^2 = \frac{16}{25} = 0.64 = 64\%$
- (c) Check normalization: $|\frac{3}{5}|^2 + |\frac{4}{5}|^2 = \frac{9}{25} + \frac{16}{25} = \frac{25}{25} = 1 \checkmark$

2. Consider the state $|\psi\rangle = a|0\rangle + b|1\rangle$ where a and b are real numbers.

- (a) If probability of $|0\rangle$ is $\frac{1}{4}$, then $|a|^2 = \frac{1}{4}$
- (b) Probability of $|1\rangle = 1 - \frac{1}{4} = \frac{3}{4}$ (probabilities must sum to 1)
- (c) If a is positive: $a = \frac{1}{2}$ and $b = \pm\frac{\sqrt{3}}{2}$ (since $|b|^2 = \frac{3}{4}$)

Quantum Interference

3. Consider these two quantum states:

$$|\psi_1\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$$|\psi_2\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

- (a) Yes, both states have identical measurement probabilities: 50% chance of $|0\rangle$ and 50% chance of $|1\rangle$. The only difference is the phase (sign) of the $|1\rangle$ component.
- (b) Adding the states: $|\psi_1\rangle + |\psi_2\rangle = \frac{2}{\sqrt{2}}|0\rangle + 0|1\rangle = \sqrt{2}|0\rangle$ After normalization: $|0\rangle$
- (c) The $|1\rangle$ components canceled out completely (+1 and -1 sum to 0). This is destructive interference.

Quantum Gates

4. Apply the following gates to the given states:

- (a) $X|0\rangle = |1\rangle$
- (b) $H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle$
- (c) $Z\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle = |-\rangle$

5. The Hadamard gate has the property that $H^2 = I$.

- (a) $|0\rangle \xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \xrightarrow{H} |0\rangle$
- (b) $|1\rangle \xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \xrightarrow{H} |1\rangle$
- (c) The Hadamard gate is its own inverse. Applying it twice returns you to the original state.

Circuit Analysis

6. Circuit analysis: $|0\rangle \rightarrow H \rightarrow Z \rightarrow H \rightarrow ?$

- (a) After first H : $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$
- (b) After Z : $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$
- (c) After second H : $|1\rangle$
- (d) This circuit implements the X gate (converts $|0\rangle$ to $|1\rangle$)

7. To convert $|1\rangle$ to $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle$, apply the Hadamard gate: $H|1\rangle = |-\rangle$

Multiple Qubits

8. Two-qubit system starting in $|00\rangle$:

- (a) Apply H to first qubit only: $(H \otimes I)|00\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$
- (b) Apply H to both qubits: $(H \otimes H)|00\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$
- (c) Probability of measuring $|11\rangle = |\frac{1}{2}|^2 = \frac{1}{4} = 25\%$

9. True or False?

- (a) False. The state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ cannot be written as $(a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle)$ for any values of a, b, c, d . This is an entangled state.
- (b) True. $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) = \left[\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\right] \otimes \left[\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\right] = |+\rangle \otimes |+\rangle$

Looking Ahead: Two-Qubit Gates

10. CNOT gate behavior:

- (a) $\text{CNOT}|00\rangle = |00\rangle$ (first qubit is 0, so second qubit unchanged)
- (b) $\text{CNOT}|10\rangle = |11\rangle$ (first qubit is 1, so second qubit flipped)
- (c) $\text{CNOT}|01\rangle = |01\rangle$ (first qubit is 0, so second qubit unchanged)
- (d) $\text{CNOT}|11\rangle = |10\rangle$ (first qubit is 1, so second qubit flipped)

11. CNOT applied to $\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$:

$$\frac{1}{\sqrt{2}}(\text{CNOT}|00\rangle + \text{CNOT}|10\rangle) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

This creates an entangled state! The qubits are now correlated - if you measure the first qubit as 0, the second will definitely be 0; if the first is 1, the second will definitely be 1.

Synthesis

12. Challenge Problem: Swap amplitudes of $|0\rangle$ and $|1\rangle$ components.

- (a) The X gate accomplishes this: $X(\alpha|0\rangle + \beta|1\rangle) = \alpha|1\rangle + \beta|0\rangle$
- (b) Input: $\frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}|1\rangle$ Expected output: $\frac{1}{\sqrt{3}}|1\rangle + \sqrt{\frac{2}{3}}|0\rangle$
- (c) Apply X gate: $X\left(\frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}|1\rangle\right) = \frac{1}{\sqrt{3}}|1\rangle + \sqrt{\frac{2}{3}}|0\rangle$ ✓